From Fourier modes to the CMB sky

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So far we have generated Gaussian random fields in one and two-dimensional Euclidean spaces. We now extend the same ideas to a field defined on the surface of a sphere.

The Cosmic Microwave Background (CMB) temperature can be described as a random field on the sky,

$$\frac{\Delta T}{T}(\theta, \phi). \tag{1}$$

On a plane, we decomposed a field into Fourier modes. On the sphere, the analogous basis is provided by the **spherical harmonics** $Y_{\ell m}(\theta, \phi)$.

The goal of this exercise is to use a theoretical CMB angular power spectrum C_ℓ to generate a simplified realization of the CMB sky.

0.1 The CMB angular power spectrum

The temperature field on the sphere can be expanded as

$$\frac{\Delta T}{T}(\theta, \phi) = \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} a_{\ell m} Y_{\ell m}(\theta, \phi). \tag{2}$$

For a statistically isotropic Gaussian CMB field,

$$\langle a_{\ell m} a^*_{\ell' m'} \rangle = C_\ell \delta_{\ell \ell'} \delta_{m m'}. \tag{3}$$

Therefore,

$$\left\langle |a_{\ell m}|^2 \right\rangle = C_\ell. \tag{4}$$

Before coding

Based on what you learned about Fourier modes and Gaussian random fields, answer the following questions:

1. What quantity in the previous exercises plays a role analogous to C_ℓ?
2. What quantities are analogous to the Fourier coefficients $\delta_{\vec{k}}$?
3. If two CMB skies are generated using exactly the same C_ℓ, should the two maps be identical?
4. What statistical properties should they share?
5. In Fourier space, large k corresponds to small spatial scales. What do you expect large ℓ to represent on the sphere?

0.2 Generate a theoretical CMB power spectrum

If possible, install and use `CAMB` to calculate a theoretical CMB temperature angular power spectrum.

Use a flat cosmological model with

$$\Omega_{\rm cdm} = 0.25, \qquad \Omega_b = 0.05, \qquad \Omega_\Lambda = 0.70, \tag{5}$$

and

$$H_0 = 70\,{\rm km\,s^{-1}\,Mpc^{-1}}. \tag{6}$$

Generate the temperature angular power spectrum

$$C_\ell^{TT}. \tag{7}$$

Plot the spectrum as a function of ℓ.
If you cannot install CAMB, use the provided file

`Cl_camb_exo.txt`

containing the multipoles ℓ and the corresponding temperature power spectrum.
Important: Check whether the quantity returned by CAMB or contained in the file is C_ℓ itself or the commonly plotted quantity

$$D_\ell = \frac{\ell(\ell+1)}{2\pi} C_\ell. \tag{8}$$

For the generation of the random coefficients below, you need C_ℓ.

0.3 Understanding spherical harmonics

Before generating a random CMB sky, explore a few individual spherical harmonics.
Use the spherical-harmonic functions available in the `scipy.special` library.
Consult the documentation for the version of SciPy installed on your computer and check carefully the convention used for the angular arguments.
Use

$$0 \leq \theta < 2\pi \tag{9}$$

for the azimuthal angle and

$$0 \leq \phi \leq \pi \tag{10}$$

for the polar angle.
Construct a two-dimensional angular grid using

`np.meshgrid()`

and visualize a few individual modes, for example

$$(\ell, m) = (1,0),\ (2,0),\ (2,1),\ (5,2). \tag{11}$$

Questions

1. How does the angular structure change when ℓ increases?
2. What changes when m is varied while keeping ℓ fixed?
3. Approximately what angular scale do you associate with a multipole ℓ?

0.4 Generate random spherical-harmonic coefficients

We will now generate a simplified Gaussian CMB realization using only multipoles

$$1 \leq \ell \leq 20. \tag{12}$$

For each multipole ℓ, generate Gaussian random coefficients $a_{\ell m}$ whose variance is determined by C_ℓ.
For $m > 0$, the real and imaginary parts of $a_{\ell m}$ are independent Gaussian random variables.
Think carefully about the variance that each component must have in order to satisfy

$$\left\langle |a_{\ell m}|^2 \right\rangle = C_\ell. \tag{13}$$

Reality condition

The CMB temperature field is real. Therefore, the spherical-harmonic coefficients cannot all be generated independently.

They must satisfy

$$a_{\ell,-m} = (-1)^m a^*_{\ell m}. \tag{14}$$

Answer before implementing this condition:

1. What is the analogous condition for the Fourier coefficients of a real field?
2. Why must $a_{\ell 0}$ be real?
3. What would happen if all positive and negative m coefficients were generated independently?

Generate the coefficients $a_{\ell m}$ satisfying Eq. 14.

0.5 Generate a CMB realization

Construct the temperature map using

$$\frac{\Delta T}{T}(\theta, \phi) = \sum_{\ell=1}^{20} \sum_{m=-\ell}^{\ell} a_{\ell m} Y_{\ell m}(\theta, \phi). \tag{15}$$

Use approximately 300 values in each angular direction.
Verify numerically that the resulting temperature map is real.
Plot the resulting map as a function of θ and ϕ.

Interpretation

Answer:

1. Why does the map contain only relatively large angular structures?
2. What would you expect to happen if you increased $\ell_{\max}$ from 20 to 100?
3. Would another random realization generated from the same C_ℓ look identical?
4. What would remain statistically the same?

0.6 Does one CMB sky reproduce the input C_ℓ?

For your generated realization, estimate the angular power at each multipole using

$$\widehat{C}_\ell = \frac{1}{2\ell + 1} \sum_{m=-\ell}^{\ell} |a_{\ell m}|^2. \tag{16}$$

Compare $\widehat{C}_\ell$ from your realization with the theoretical input C_ℓ.
Answer:

1. Does a single realization reproduce the theoretical C_ℓ exactly?
2. At a fixed ℓ, how many m modes contribute to the estimate?
3. Is the scatter larger at low or high ℓ?
4. Why does increasing ℓ provide more independent modes with which to estimate C_ℓ?

Generate several independent CMB realizations and compare their $\widehat{C}_\ell$.
Discuss what happens when you average the estimated spectra over several realizations.

0.7 From a rectangular plot to a sphere

The (θ, ϕ) representation of the map is a rectangular representation of a field that actually lives on a sphere.

To visualize the geometry more directly, you may construct the Cartesian coordinates

$$x = \sin\phi\cos\theta, \tag{17}$$
$$y = \sin\phi\sin\theta, \tag{18}$$
$$z = \cos\phi, \tag{19}$$

and use the temperature fluctuation as the color of the spherical surface.

Compare the rectangular representation with the spherical visualization.

Explain why structures close to the poles appear distorted in the rectangular (θ, ϕ) representation.

0.8 A controlled experiment

Repeat the map generation using different maximum multipoles, for example

$$\ell_{\max} = 5, \qquad 10, \qquad 20. \tag{20}$$

Use the same random coefficients for the modes common to all three maps.

Before plotting the maps, predict how their appearance will change as $\ell_{\max}$ increases.

Then compare the three maps and explain your observations in terms of angular scales.

Connection with the previous practical

Complete the following correspondence:

Flat space	**Sphere**
Fourier modes	Spherical harmonics
k	ℓ
δ_k	$a_{\ell m}$
$P(k)$	C_ℓ
small k	low ℓ
large k	high ℓ

In your own words, explain why generating a Gaussian CMB sky from C_ℓ is conceptually similar to generating the Gaussian random fields from $P(k)$ in the previous exercise.

Individual challenge

During the practical session, the instructor may ask you to modify one aspect of your CMB realization.

You will first predict the result without running your code, and then test your prediction.

Possible modifications include:

- changing $\ell_{\max}$;
- removing a selected range of multipoles;
- changing only the random realization while keeping C_ℓ fixed;
- artificially increasing the power in a selected range of ℓ;
- identifying a violation of the reality condition;
- explaining an unexpected imaginary component in the temperature map.

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